

## RESEARCH PAPER

# Long-term outcome of hemispheric surgery at different ages in 61 epilepsy patients

Anita Althausen,<sup>1</sup> Ulrike Gleissner,<sup>2</sup> Christian Hoppe,<sup>1</sup> Robert Sassen,<sup>1</sup> Svenja Buddewig,<sup>1</sup> Marec von Lehe,<sup>3</sup> Johannes Schramm,<sup>3</sup> Christian Erich Elger,<sup>1</sup> Christoph Helmstaedter<sup>1</sup>

<sup>1</sup>Department of Epileptology, University of Bonn Medical Centre, Bonn, Germany

<sup>2</sup>Department of Pediatric Neurology, LVR-Clinic Bonn, Bonn, Germany

<sup>3</sup>Department of Neurosurgery, University of Bonn Medical Centre, Bonn, Germany

## Correspondence to

Anita Althausen, Department of Epileptology, University of Bonn Medical Centre, Sigmund-Freud Str. 25, 53105 Bonn, Germany; anita.althausen@ukb.uni-bonn.de

Received 1 August 2012

Revised 29 November 2012

Accepted 2 December 2012

Published Online First

25 December 2012

## ABSTRACT

**Objective** Hemispheric neurosurgery is an established treatment for severe epilepsy caused by extended unilateral brain pathology. However, it is still an unresolved question at which age surgery should best be performed. In light of decreasing plasticity and the cumulative impact of seizures and anticonvulsants on neurodevelopment, early surgery appears preferable.

**Methods** We retrospectively investigated the medical, cognitive-behavioural and psychosocial long-term outcome (follow-up: 9.4 years (1.1–19.4)) of hemispherectomy as a function of age at surgery (early: <7 years/intermediate: 7–16 years/late: >16 years) based on a structured postal survey in a large patient sample (N=61/81, return rate: 75%).

**Results** At follow-up, 45 (74%) patients were seizure free. Presurgical levels of intelligence were below average in most patients (79%) and postsurgical cognition largely resembled the presurgical capacities. Best seizure outcome was obtained for early surgery patients (90% seizure free). Patients with late surgery, however, exhibited higher presurgical and postsurgical intelligence and better psychosocial achievements. Binary logistic regression identified better presurgical intelligence and higher age at surgery as positive predictors of postsurgical intelligence. Lower presurgical intelligence and postsurgical seizure freedom predicted intellectual pre-post improvements.

**Conclusions** Our data confirm the efficacy and cognitive safety of hemispheric surgeries performed at different ages. Eligible older and high functioning patients can be perfect candidates. Presurgical intelligence serves as indicator of the functional integrity of the contralateral hemisphere, which mainly determines postsurgical cognition and psychosocial outcome. Seizure freedom promotes cognitive improvement. As many unknown factors determined age at surgery, our retrospective data neither question early surgeries nor suggest postponing surgery.

## INTRODUCTION

Hemispherectomy techniques play a crucial role for the treatment of refractory epilepsies involving global unilateral damage.<sup>1–7</sup> Typical aetiologies comprise acquired vascular insults, neurodevelopmental disorders and progressive diseases such as Rasmussen's encephalitis. Hemispheric surgery eliminates seizures in 43–90% of patients, with disconnective procedures being equally effective as

resective hemispherectomy combined with minimised risks.<sup>8,9</sup>

Preoperative intelligence (IQ) or developmental quotients in candidates for hemispherectomy are usually well below average and most studies yielded no or minor cognitive changes after surgery.<sup>10–16</sup> However, reports have rarely gone beyond the evaluation of seizures and sometimes intelligence, while the psychosocial long-term outcome, including academic and job achievements, was seldom assessed. Furthermore, due to small sample sizes, no clinical outcome prediction model could yet be established. In particular, the optimal timing of surgery with regard to neurocognitive development, despite being a pressing clinical issue, remains uncertain so far. General neuropsychological considerations suggest that younger patients might benefit from the maturing brain's higher plasticity and an earlier release from the negative impact of epilepsy (ie, seizures, antiepileptic drugs) on brain development,<sup>17–19</sup> and several authors have argued in favour of early surgical intervention and short epilepsy duration for hemispheric surgery patients.<sup>7,10,14,15,20–24</sup>

Here we report the combined seizure, cognitive-behavioural and psychosocial long-term outcome of one of the largest series of patients who underwent hemispherectomy for the treatment of refractory epilepsy. We were particularly interested in the role of age at surgery in relation to presurgical and postsurgical neurocognitive development.

## MATERIALS AND METHODS

### Patients

From 1988 to 2007, 110 patients underwent hemispheric surgery for medically refractory epilepsy at our department. Inclusion criteria were a minimum age of 6 years at follow-up, a postsurgical interval of at least 1 year, and residence in Germany for postal availability. Of 86 eligible patients, five could not be located.

To evaluate the effect of age at surgery, the sample was divided into three subgroups based on neurodevelopmental considerations.<sup>18,25</sup> The early surgery group (ES) included patients younger than 7 years at surgery whose brain and language development was still in progress; the intermediate surgery group (IS) consisted of patients operated on between 7–16 years of age whose brain was still in maturation while language development was largely complete; and the late surgery group (LS) comprised patients

**To cite:** Althausen A, Gleissner U, Hoppe C, et al. *J Neurol Neurosurg Psychiatry* 2013;**84**:529–536.

older than 16 years at surgery whose brain was essentially mature. Age at surgery was not under experimental control for obvious ethical and practical reasons. Therefore, this study can only evaluate outcome in conjunction with the established clinical procedures.

## Measures

Postsurgical outcome was assessed by means of a newly constructed 57-item questionnaire (available upon request) requiring parental ratings on three predefined dimensions: seizure outcome, cognitive-behavioural performance and psychosocial (including socioeconomic) achievement. Furthermore, information about early development, supportive educative activities, school career, professional career, functional independence, relationship status and postsurgical behavioural changes were ascertained. Questionnaires were submitted via postal mailing with explicit request to participate even if the postsurgical results were not satisfying. A small material reward was granted upon participation. Additional data from the patient records were considered with permission of the patients/parents. Written informed consent was obtained from all participating parents/patients.

## Statistical analysis

Group differences were analysed by  $\chi^2$  or Kruskal-Wallis tests where appropriate. To delimitate cells with expected values less than 5, variables with more than three levels were reduced to 2-point or 3-point scaling for  $\chi^2$ -analysis. If expected values remained below five for more than 30% of cells, results were rechecked by Fisher's exact test. Pre-post changes in intelligence ratings were tested by Wilcoxon signed-rank test. Correlations were calculated as Spearman's  $r$ . To identify clinical predictors of the long-term outcome, stepwise binary logistic regression analyses were conducted separately for seizure outcome (seizure free yes/no), intellectual outcome (average/below average), postsurgical intellectual deterioration (risk) and improvement (benefit). Age at surgery, side of surgery, aetiology (acquired, developmental, progressive), and preoperative intelligence were considered as factors in all analyses. Age at epilepsy onset, duration of epilepsy and current seizure status were included if bivariate correlations indicated potential relevance. Goodness-of-fit is reported as Pseudo- $R^2$  (Nagelkerke). All analyses were performed with SPSS (V.18.0.0, German version) at a significance level of 0.05 (two-tailed).

## RESULTS

### Patient characteristics

Completed questionnaires were obtained from 61/81 patients/families (75%). Patient characteristics are shown in table 1. The mean postsurgical follow-up was 9.4 years (median 7.6; SD 5.4; range 1.1–19.4). For 10 patients (16%), surgery was performed as described by Rasmussen,<sup>1</sup> 41 underwent transsylvian-transventricular functional hemispherectomy,<sup>4</sup> 6 transcortical hemispherectomy<sup>2</sup> and 4 modified hemispherectomies involving incomplete disconnection. Aetiology was classified as acquired (vascular insults/porencephaly: N=36), developmental (hemimegalencephaly: 6, cortical dysplasia: 1, other congenital malformations: 3), progressive (encephalitis: 8, Sturge-Weber syndrome: 3, hemiconvulsion-hemiplegia-epilepsy syndrome: 1) or unclear (multiple aetiology: 1, unknown: 2). Presurgical intelligence was estimated as below average for the majority of patients (45/57, 79%). Eleven patients (19%) were rated as having had average intelligence, one patient as above average.

Patient groups (ES/IS/LS N=19/22/20) did not differ as regards gender, side of surgery, hemispherectomy technique and follow-up interval. However, ES patients experienced presurgical seizures for a shorter time and were younger currently and at epilepsy onset. As a non-significant trend, ES comprised the lowest proportion of patients with acquired ( $p=0.050$ ) and the highest with developmental pathology ( $p=0.093$ ). Furthermore, a higher percentage of ES patients had speech and motor retardation in early childhood and, correspondingly, participated in more supportive educative activities. LS patients received higher ratings of presurgical intelligence than both other groups (3-point scale; ES-LS:  $p=0.017$ ; IS-LS:  $p=0.022$ ) and developmental delay at kindergarten age was less frequent for this subgroup (ES-LS:  $p=0.008$ ; IS-LS:  $p=0.016$ ).

### Seizure outcome

At follow-up, 45 (74%) patients were seizure free. Seizures improved for all remaining patients (table 2). Most seizure-free patients were off anticonvulsants (36/45, 80%), compared with 1/16 (6%) patients with continuing seizures ( $p<0.001$ ). Overall, 37 patients (61%) were no longer on antiepileptic drugs (AEDs). A higher seizure remission rate (90% vs 60%,  $p=0.035$ ) and a higher proportion of patients with no further need for anticonvulsants ( $p=0.034$ ) was reported for ES as compared with LS patients (figure 1).

### Cognitive-behavioural outcome

Current intellectual levels and cognitive-behavioural changes are shown in table 2 and illustrated in figures 2 and 3. Current intelligence was estimated as below average for 41/58 patients (71%), with 17 (29%) rated as functioning on an average intellectual level and none above average. Nevertheless, current intelligence improved significantly compared with the preoperative ratings (Wilcoxon;  $p=0.010$ ). For 21 patients (38%), intelligence was estimated as better than at the time of surgery, worse for 5 (9%), and unchanged for 29 patients (53%). Improvement in at least one cognitive domain was indicated for 52 patients (85%). Overall, patients improved in a mean number of 3.0 cognitive domains (SD 2.5; range 0–8), with improvements mainly concerning attention (72%), expressive speech (46%), language comprehension (41%) and memory (41%). Behavioural improvements were reported for about half of the patients (57%).

Age at surgery was significantly related to postsurgical intelligence and cognitive-behavioural development. LS patients were best rated in current intelligence (3-point scale,  $p=0.006$ ). However, in contrast to the presurgical ratings, postsurgical intelligence was not significantly different for IS and LS patients ( $p=0.085$ ). More ES than IS and LS patients were estimated as mentally retarded postoperatively (ES-LS:  $p=0.001$ ; ES-IS:  $p=0.044$ ).

Age at surgery had no effect on pre-post intelligence improvements. Within groups, a significant intellectual pre-post improvement was found for IS (Wilcoxon,  $p=0.015$ ) and LS patients (Wilcoxon,  $p=0.046$ ). IS patients improved in a higher number of cognitive domains than LS patients ( $p=0.009$ ) and exhibited behavioural improvements more often than ES patients ( $p=0.014$ ).

### Psychosocial outcome

Table 3 summarises the psychosocial outcome. At follow-up, 60% of the eligible patients (current age >16 years, N=41) had not attended regular educational programmes at secondary schools and 55% did not achieve regular graduation. One patient qualified for university admission; none pursued

**Table 1** Presurgical patient characteristics

|                                                         | Total     | Age at surgery |              |          | p Value | Group comparison |                  |                  |
|---------------------------------------------------------|-----------|----------------|--------------|----------|---------|------------------|------------------|------------------|
|                                                         |           | Early          | Intermediate | Late     |         | ES-LS<br>p Value | ES-IS<br>p Value | IS-LS<br>p Value |
| Gender (m/f) (N=61)                                     | 29/32     | 8/11           | 11/11        | 10/10    | 0.849*  | 0.621*           | 0.613*           | 1.00*            |
| Present age (years) m/SD (N=61)                         | 24/12.2   | 13.8/5.8       | 19.9/5.8     | 38.1/8.3 | <0.001† | <0.001†          | 0.005†           | <0.001†          |
| Age at surgery (years) m/SD (N=61)                      | 14.5/12.0 | 3.1/2.3        | 11.1/3.0     | 29.2/8.3 |         |                  |                  |                  |
| Age at epilepsy onset (years) m/SD (N=59)               | 2.8/3.7   | 0.8/1.0        | 2.6/2.5      | 4.9/5.2  | 0.002†  | 0.001†           | 0.009†           | 0.140†           |
| Duration of epilepsy before surgery (years) m/SD (N=59) | 11.9/10.8 | 2.3/2.1        | 8.8/3.4      | 24.3/8.7 | <0.001† | <0.001†          | <0.001†          | <0.001†          |
| Follow-up interval (years) m/SD (N=61)                  | 9.4/5.4   | 10.6/5.3       | 8.8/5.4      | 8.9/5.5  | 0.408†  | 0.261†           | 0.229†           | 0.940†           |
| Side of surgery (right/left) (N=61)                     | 27/34     | 9/10           | 9/13         | 9/11     | 0.914*  | 0.882*           | 0.678*           | 0.789*           |
| Hemispherectomy technique (N=61)                        |           |                |              |          |         |                  |                  |                  |
| Rasmussen                                               | 10 (16%)  | 4 (21%)        | 3 (14%)      | 3 (15%)  | 0.892*  | 0.705*           | 0.782*           | 0.824*           |
| Transsylvian                                            | 41 (67%)  | 11 (58%)       | 15 (68%)     | 15 (75%) |         |                  |                  |                  |
| Transcortical                                           | 6 (10%)   | 2 (10.5%)      | 3 (14%)      | 1 (5%)   |         |                  |                  |                  |
| Modified                                                | 4 (7%)    | 2 (10.5%)      | 1 (5%)       | 1 (5%)   |         |                  |                  |                  |
| Aetiology (N=61)                                        |           |                |              |          |         |                  |                  |                  |
| Acquired                                                | 36 (59%)  | 7 (37%)        | 15 (68%)     | 14 (70%) | 0.142*  | 0.090*           | 0.090*           | 0.962*           |
| Progressive                                             | 12 (20%)  | 5 (26%)        | 4 (18%)      | 3 (15%)  |         |                  |                  |                  |
| Developmental                                           | 10 (16%)  | 6 (32%)        | 2 (9%)       | 2 (10%)  |         |                  |                  |                  |
| Unclear†                                                | 3 (5%)    | 1 (5%)         | 1 (5%)       | 1 (5%)   |         |                  |                  |                  |
| Preoperative intelligence (N=57)                        |           |                |              |          |         |                  |                  |                  |
| Severe retardation                                      | 11 (19%)  | 3 (19%)        | 7 (32%)      | 1 (5%)   | 0.118*§ | 0.017*§          | 0.944*§          | 0.022*§          |
| Moderate retardation                                    | 18 (32%)  | 8 (50%)        | 7 (32%)      | 3 (16%)  |         |                  |                  |                  |
| Mild retardation                                        | 16 (28%)  | 3 (19%)        | 5 (23%)      | 8 (42%)  |         |                  |                  |                  |
| Average                                                 | 11 (19%)  | 1 (6%)         | 3 (14%)      | 7 (37%)  |         |                  |                  |                  |
| Above average                                           | 1 (2%)    | 1 (6%)         | 0 (0%)       | 0 (0%)   |         |                  |                  |                  |
| Supportive activities early childhood m/SD (N=59)       | 2.7/1.7   | 4.0/1.3        | 2.6/1.2      | 1.5/1.4  | <0.001† | <0.001†          | 0.001†           | 0.005†           |
| Supportive activities elementary school m/SD (N=52)     | 1.4/1.2   | 1.9/1.2        | 1.3/0.9      | 0.8/1.2  | 0.034†  | 0.017†           | 0.126†           | 0.114†           |
| Motor retardation ≤3 years (N=59)                       | 41 (70%)  | 17 (90%)       | 13 (62%)     | 11 (58%) | 0.069*  | 0.027*           | 0.044*           | 0.076*           |
| Speech retardation ≤3 years (N=59)                      | 35 (59%)  | 18 (95%)       | 11 (52%)     | 6 (32%)  | <0.001* | <0.001*          | 0.003*           | 0.184*           |
| Behavioural problems ≤3 years (N=53)                    | 15 (28%)  | 4 (31%)        | 6 (29%)      | 5 (26%)  | 0.962*  | 0.783*           | 0.891*           | 0.873*           |
| Developmental delay in kindergarten (N=49)              | 23 (43%)  | 10 (59%)       | 11 (52%)     | 2 (13%)  | 0.020*  | 0.008*           | 0.691*           | 0.016*           |

\*Pearson's  $\chi^2$ .

†Kruskal-Wallis test.

‡Excluded from analysis.

§3-point scale: intelligence (average/mild retardation/moderate or severe retardation).

ES, early surgery; IS, intermediate surgery; LS, late surgery.

academic studies. For 40% of patients operated on at school age (6–19 years; N=30), postsurgical improvement in school performance was reported. Out of the now adult patients (current age >19 years; N=35), more than 70% had not completed a vocational training, had no regular job and had a monthly income of less than €400; only 41% lived without assistance, and only 20% were currently in a relationship. Out of all patients, between 20% and 61% were rated as independent in individual daily activities. Fourteen patients (23%) required permanent attendance.

In accordance with their higher estimated intelligence, more LS than ES patients attended regular secondary schools ( $p=0.008$ ), achieved graduation ( $p=0.002$ ) and managed to live independently ( $p=0.028$ ). None of the currently adult ES patients were employed on a regular basis as compared with six LS patients (30%).

### Outcome prediction

Postoperative seizure cessation correlated with younger age at surgery ( $r=0.304$ ;  $p=0.017$ ) and at follow-up ( $r=0.262$ ;  $p=0.041$ ), with pre-post intelligence improvement ( $r=0.349$ ;  $p=0.009$ ), and, as a non-significant trend, with younger age at epilepsy onset ( $r=0.254$ ;  $p=0.052$ ), shorter duration of epilepsy

before surgery ( $r=0.227$ ;  $p=0.083$ ), and a higher number of improved cognitive domains ( $r=0.246$ ;  $p=0.056$ ). Stepwise logistic regression identified age at surgery as the only significant predictor of postsurgical seizure control ( $p=0.002$ ;  $R^2=0.235$ ), with greater odds of postoperative seizure freedom for patients operated at a younger age (OR 0.92, 95% CI 0.86 to 0.97,  $p=0.005$ ).

Current intellectual levels correlated positively with preoperative intelligence ( $r=0.566$ ;  $p<0.001$ ), age at surgery ( $r=0.466$ ;  $p<0.001$ ), age at epilepsy onset ( $r=0.391$ ;  $p=0.003$ ) and duration of epilepsy before surgery ( $r=0.407$ ;  $p=0.002$ ); negatively with early behavioural problems ( $r=-0.308$ ;  $p=0.028$ ), speech retardation ( $r=-0.312$ ;  $p=0.018$ ), developmental delay ( $r=-0.311$ ;  $p=0.026$ ) and the number of supportive activities during early childhood ( $r=-0.276$ ;  $p=0.036$ ) and elementary school ( $r=-0.432$ ;  $p=0.014$ ). Regression modelling confirmed preoperative intelligence and age at surgery as predictors of the intellectual outcome ( $p<0.0001$ ;  $R^2=0.508$ ). The odds of average postsurgical intelligence were greater for patients with average than below-average intelligence before surgery (OR 21.38, 95% CI 3.26 to 140.39,  $p=0.001$ ) and increased with each additional year of age at surgery (OR 1.08, 95% CI 1.01 to 1.16,  $p=0.036$ ).

**Table 2** Seizures, AEDs and cognitive-behavioural outcome

|                                                     | Total    | Age at surgery |              |          | p Value | Group comparison |                  |                  |
|-----------------------------------------------------|----------|----------------|--------------|----------|---------|------------------|------------------|------------------|
|                                                     |          | Early          | Intermediate | Late     |         | ES-LS<br>p Value | ES-IS<br>p Value | IS-LS<br>p Value |
| Seizure outcome (N=61)                              |          |                |              |          |         |                  |                  |                  |
| Seizure free                                        | 45 (74%) | 17 (89.5%)     | 16 (73%)     | 12 (60%) | 0.111*† | 0.035*†          | 0.177*†          | 0.382*†          |
| Significant reduction                               | 11 (18%) | 2 (10.5%)      | 3 (14%)      | 6 (30%)  |         |                  |                  |                  |
| Some reduction                                      | 5 (8%)   | 0 (0%)         | 3 (14%)      | 2 (10%)  |         |                  |                  |                  |
| Unchanged/worse                                     | 0 (0%)   | 0 (0%)         | 0 (0%)       | 0 (0%)   |         |                  |                  |                  |
| No more AEDs (N=57)                                 | 37 (61%) | 14 (74%)       | 15 (68%)     | 8 (40%)  | 0.066*  | 0.034*           | 0.699*           | 0.067*           |
| Postoperative intelligence (N=58)                   |          |                |              |          |         |                  |                  |                  |
| Severe retardation                                  | 4 (7%)   | 1 (5%)         | 2 (10%)      | 1 (5%)   | 0.006*‡ | 0.002*‡          | 0.085*‡          | 0.265*‡          |
| Moderate retardation                                | 18 (31%) | 11 (58%)       | 5 (25%)      | 2 (11%)  |         |                  |                  |                  |
| Mild retardation                                    | 19 (33%) | 6 (32%)        | 7 (35%)      | 6 (32%)  |         |                  |                  |                  |
| Average                                             | 17 (29%) | 1 (5%)         | 6 (30%)      | 10 (53%) |         |                  |                  |                  |
| Above average                                       | 0 (0%)   | 0 (0%)         | 0 (0%)       | 0 (0%)   |         |                  |                  |                  |
| Intelligence change (N=55)                          |          |                |              |          |         |                  |                  |                  |
| Worse                                               | 5 (9%)   | 3 (19%)        | 2 (10%)      | 0 (0%)   | 0.050*  | 0.024*           | 0.749*           | 0.035*           |
| Unchanged                                           | 29 (53%) | 6 (38%)        | 8 (40%)      | 15 (79%) |         |                  |                  |                  |
| Better                                              | 21 (38%) | 7 (44%)        | 10 (50%)     | 4 (21%)  |         |                  |                  |                  |
| Intellectual deterioration: worse/not worse§ (N=55) | 5/50     | 3/13           | 2/18         | 0/19     | 0.155*† | 0.048*†          | 0.451*†          | 0.157*†          |
| Intellectual improvement: better/not better§ (N=55) | 21/34    | 7/9            | 10/10        | 4/15     | 0.153*† | 0.150*†          | 0.709*†          | 0.060*†          |
| Improved cognitive domains (max. 9) m/SD (N=58)     | 3.0/2.5  | 3.1/2.1        | 4.0/2.5      | 2.0/2.4  | 0.016¶  | 0.044¶           | 0.219¶           | 0.009¶           |
| Behavioural improvement (N=58)                      | 33 (57%) | 6 (33%)        | 16 (73%)     | 11 (61%) | 0.040*  | 0.100*           | 0.014*           | 0.441*           |

\*Pearson's  $\chi^2$ .

†2-point scale: seizure status (seizure free/not seizure free); intellectual deterioration (worse/unchanged or better); intellectual improvement (better/unchanged or worse).

‡3-point scale: intelligence (average/mild retardation/moderate or severe retardation).

§Derived from intelligence change.

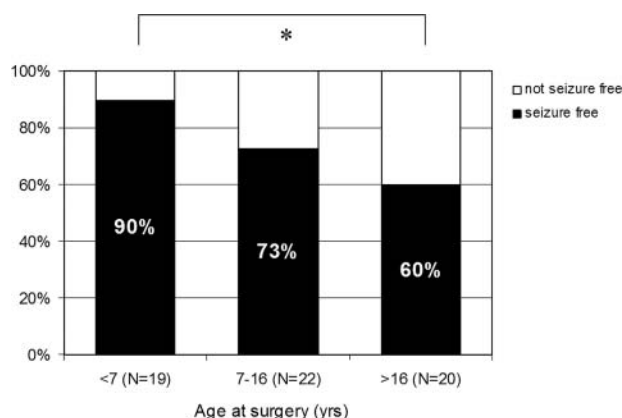
¶Kruskal-Wallis test.

ES, early surgery; IS, intermediate surgery; LS, late surgery.

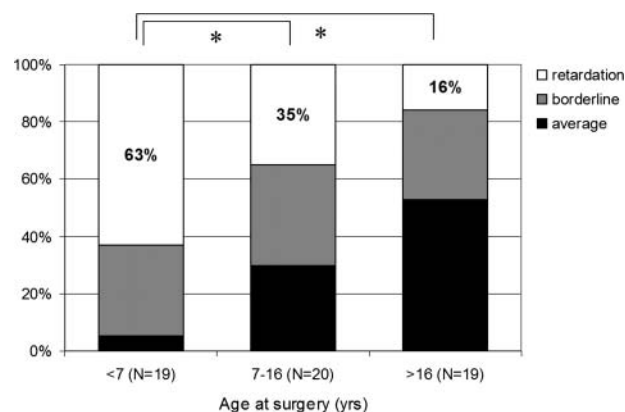
Cognitive benefit in terms of pre-post improvement in intelligence was associated with seizure freedom ( $r=0.349$ ;  $p=0.009$ ), a higher number of improved cognitive domains ( $r=0.576$ ;  $p<0.001$ ); lower presurgical intellectual levels ( $r=-0.474$ ;  $p<0.001$ ); and more supportive activities in early childhood ( $r=0.284$ ;  $p=0.036$ ). Regression analysis indicated postsurgical seizure cessation and preoperative intelligence as decisive for pre-post intellectual improvement ( $p<0.0001$ ;  $R^2=0.463$ ), with higher odds of intellectual benefit for patients with presurgical

lower levels of functioning (OR 0.27; 95% CI 0.12 to 0.60,  $p=0.001$ ) and postsurgically seizure free patients (OR 17.41, 95% CI 1.67 to 181.13,  $p=0.017$ ). Only one patient with continuing seizures (1/16, 6%) improved in estimated intelligence as compared with 20/45 (44%) seizure-free patients.

Cognitive risk in terms of postsurgically decreased intelligence correlated with better presurgical intelligence ( $r=0.351$ ;  $p=0.009$ ) and younger age at surgery ( $r=-0.295$ ;  $p=0.029$ ). Logistic regression analysis confirmed both variables as



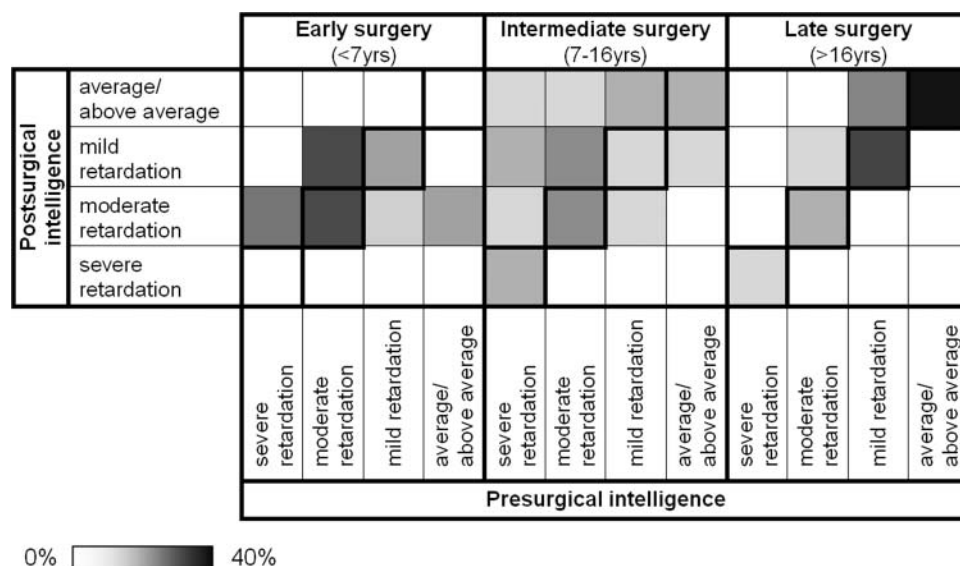
**Figure 1** Seizure outcome by age at surgery. Seizure remission rates after a mean postsurgical interval of 9 years (SD 5) displayed separately for patients with early (<7 years), intermediate (7–16 years) and late (>16 years) surgery.



**Figure 2** Cognitive outcome by age at surgery. Parental ratings of the patients' current intellectual level displayed separately for patients with early (<7 years), intermediate (7–16 years) and late (>16 years) surgery. Retardation: moderate or severe retardation, borderline: mild retardation, average: average intelligence.



**Figure 3** Pre-post changes in intelligence. Preoperative and postoperative (ie, at follow-up) intelligence ratings per cent displayed as intensity on a grayscale from 0% =white to 40%=black.



predictors of intellectual deterioration ( $p < 0.0001$ ;  $R^2 = 0.699$ ), with greater risk for patients with higher presurgical intelligence (OR 21.30, 95% CI 1.56 to 291.49,  $p = 0.022$ ) and early surgery (OR 0.67, 95% CI 0.43 to 1.05,  $p = 0.078$ ).

Cognitive-behavioural and psychosocial outcomes were highly correlated. Current intelligence ratings correlated positively with capacities for regular schooling ( $r = 0.705$ ;  $p < 0.001$ ), graduation ( $r = 0.678$ ;  $p < 0.001$ ), employment ( $r = 0.415$ ,  $p = 0.022$ ), improved school performance ( $r = 0.424$ ,  $p = 0.025$ ,  $N = 28$ ), a monthly income of at least €400 ( $r = 0.422$ ,  $p = 0.020$ ,  $N = 30$ ) and the degree of functional independence ( $r = 0.402$ ,  $p = 0.010$ ,

$N = 40$ ). Postsurgical intellectual improvement correlated with improved school performance ( $r = 0.381$ ,  $p = 0.045$ ,  $N = 28$ ). Positive behavioural change was associated with improvement in cognitive domains ( $r = 0.350$ ;  $p = 0.007$ ).

#### Other clinical factors

No significant effect of aetiology, hemispherectomy technique or completeness of hemispheric disconnection on any measure of postoperative outcome was obtained (all  $p > 0.1$ ). Laterality of surgery had an effect only on personal living arrangements:

**Table 3** Psychosocial outcome

|                                                   | Total    | Age at surgery |              |          | p Value | Group comparison |         |         |
|---------------------------------------------------|----------|----------------|--------------|----------|---------|------------------|---------|---------|
|                                                   |          | Early          | Intermediate | Late     |         | ES-LS            | ES-IS   | IS-LS   |
|                                                   |          |                |              |          |         | p Value          | p Value | p Value |
| Improved school performance ( $N = 30$ )*         | 12 (40%) | 1 (25%)        | 10 (45.5%)   | 1 (25%)  | 0.600†  | 0.446†           | 0.446†  | 1.00†   |
| Secondary school: regular education ( $N = 40$ )‡ | 16 (40%) | 0 (0%)         | 5 (36%)      | 11 (58%) | 0.026†  | 0.008†           | 0.070†  | 0.208†  |
| School graduation ( $N = 40$ )‡                   |          |                |              |          |         |                  |         |         |
| Graduation                                        | 18 (45%) | 0 (0%)         | 5 (36%)      | 13 (68%) | 0.005†§ | 0.002†§          | 0.070†§ | 0.062†§ |
| Special school degree                             | 10 (25%) | 2 (29%)        | 5 (36%)      | 3 (16%)  |         |                  |         |         |
| No graduation                                     | 12 (30%) | 5 (71%)        | 4 (29%)      | 3 (16%)  |         |                  |         |         |
| Vocational training ( $N = 33$ )¶                 | 9 (27%)  | 1 (20%)        | 0 (0%)       | 8 (42%)  | 0.060†  | 0.364†           | 0.164†  | 0.021†  |
| Independent living ( $N = 34$ )¶                  | 14 (41%) | 0 (0%)         | 2 (20%)      | 12 (60%) | 0.023†  | 0.028†           | 0.334†  | 0.038†  |
| Employment ( $N = 33$ )¶                          |          |                |              |          |         |                  |         |         |
| Yes                                               | 7 (21%)  | 0 (0%)         | 1 (12.5%)    | 6 (30%)  | 0.028†  | 0.125†           | 0.031†  | 0.093†  |
| Sheltered                                         | 9 (27%)  | 0 (0%)         | 5 (62.5%)    | 4 (20%)  |         |                  |         |         |
| No                                                | 17 (52%) | 5 (100%)       | 2 (25%)      | 10 (50%) |         |                  |         |         |
| Income before tax $\geq$ €400/month ( $N = 33$ )¶ | 8 (24%)  | 0 (0%)         | 1 (12.5%)    | 7 (35%)  | 0.177†  | 0.119†           | 0.411†  | 0.234†  |
| Marital status ( $N = 35$ )¶                      |          |                |              |          |         |                  |         |         |
| Married                                           | 2 (6%)   | 0 (0%)         | 0 (0%)       | 2 (10%)  | 0.458†§ | 0.211†§          | 0.283†§ | 0.760†§ |
| In relationship                                   | 5 (14%)  | 0 (0%)         | 2 (20%)      | 3 (15%)  |         |                  |         |         |
| Single                                            | 28 (80%) | 5 (100%)       | 8 (80%)      | 15 (75%) |         |                  |         |         |

\*Patients operated on at school-age (6–19 years;  $N = 30$ ).

†Pearson's  $\chi^2$ .

‡Patients older than 16 years at follow-up ( $N = 40$ ; ES/IS/LS: 7/14/20).

§2-point scale: school graduation (regular graduation/no graduation or special school degree), marital status (in relationship or married/single).

¶Patients older than 19 years at follow-up ( $N = 35$ ; ES/IS/LS: 5/10/20).

ES, early surgery; IS, intermediate surgery; LS, late surgery.

more patients with right-sided surgery lived independently (right/left: 60%/26%;  $p=0.048$ ).

## DISCUSSION

We provided comprehensive data on the seizure, cognitive-behavioural and psychosocial long-term outcome (mean: 9.4 years) from one of the largest series of epilepsy patients who underwent hemispheric surgery at different ages at a single institution.

### Seizure outcome

With 74% seizure-free patients and reduced seizure frequency for all remaining patients, seizure outcome was favourable and compares well with other long-term evaluations.<sup>13 14 22–23 26</sup> Younger age at surgery was a positive predictor of long-term seizure freedom but the correlation was only moderate. The finding was nevertheless unexpected as younger age at epilepsy onset, a lower presurgical cognitive level, more supportive training activities in early childhood and the readiness for an early surgical intervention indicate earlier and more severe brain insults with stronger involvement of the contralateral hemisphere in ES patients. Therefore, the advantageous seizure outcome in younger patients provides indirect evidence for the concept of epileptogenic network formation in previously non-epileptogenic tissue during the course of uncontrolled epilepsy.<sup>27</sup> Studies on AED withdrawal after successful focal epilepsy surgery similarly suggest a higher rate of seizure freedom with no further need for AEDs in younger patients with shorter epilepsy duration as compared with adult patients.<sup>28</sup> Evidently, the obtained age-at-surgery effect on seizures requires confirmation from other groups. In fact, a recent evaluation of seizure outcome based on International League against Epilepsy ratings for a different subgroup of our LS patients ( $N=27$ ) yielded a slightly better seizure freedom rate than obtained in this study (81%).<sup>29</sup> Differences in outcome between these evaluations can be easily explained by factors such as inclusion of incompletely disconnected cases, outcome assessment at different time points and differences in memory and interpretation when assessment is done with cognitively impaired patients versus unimpaired parents.

### Cognitive and psychosocial outcome

Consistent with other studies, the majority of patients (79%) already showed mental retardation before surgery and postsurgical functioning largely resembled the presurgical functional status.<sup>10–16 22 30</sup> Accordingly, the overall prospects remained limited and most patients were not able to succeed in regular classes, complete a vocational training or work in a regular job. In addition, functional independence remained profoundly constricted as more than half of the patients required assistance in specific daily activities and lived in assisted settings, and one third required permanent attendance. This corresponds with other reports of limited functional independence<sup>23 31</sup> and mainstream school education attainable only for a minority.<sup>3 15 24</sup> Thus, on a group level, developmental or intellectual catch-up to healthy peers after hemispherectomy cannot normally be expected.

Nevertheless, most patients benefitted in terms of cognitive-behavioural improvements in individual domains, which is undoubtedly relevant for daily life and parental assistance requirements. Improvements in intelligence were more likely for patients with presurgical below-average intelligence whereas decrements were more likely for patients with normal or only mildly impaired intelligence. We suppose that this change

pattern can at least partially be attributed to a floor/ceiling effect and little differentiation on the upper end of the intelligence scale. Of note, even among patients with no or mild presurgical intellectual impairment, more than 80% appeared improved or unchanged in intelligence following surgery. Overall, cognitive deterioration was reported for only five patients (9%), indicating low risk of adverse cognitive effects. Therefore, in line with other evidence,<sup>9–12 14 22 32</sup> our findings confirm the overall safety of hemispheric surgery as regards cognitive functioning.

Postsurgical cessation of seizures was the second predictor of cognitive improvement, with improvements in intelligence almost exclusively occurring in case of postsurgical seizure freedom. This highlights the crucial significance of seizure freedom for positive cognitive change.<sup>21 22 32</sup> In addition, the reduction or withdrawal of AEDs in seizure-free patients might have contributed to subsequent positive cognitive development.<sup>19</sup>

### Age at surgery and postsurgical outcome

The optimal timing of surgical interventions in refractory childhood epilepsies with extended unilateral brain lesions is still a pressing clinical issue but experimental studies on the role of age at surgery are unfeasible for ethical reasons. Several authors have advocated early hemispheric surgery and short epilepsy duration and some evidence seems to suggest advantages with regard to postoperative development and adaptive functioning.<sup>7 10 14–15 20–24 33</sup> We found a slightly better seizure outcome in ES patients but reported better intellectual, academic and psychosocial outcomes for LS patients. Higher age at surgery was identified as a negative predictor of the seizure outcome but a positive predictor of the cognitive outcome (besides the presurgical cognitive level). Advantageous cognitive and psychosocial outcomes in hemispherectomy patients operated at a higher age have been previously reported.<sup>29 33–38</sup> For example, Cuckiart *et al*<sup>33</sup> found no cognitive decline in any of the 14 adult hemispherectomy patients. Chiricozzi *et al*<sup>36</sup> described a case of remarkable cognitive improvement after hemispherectomy at age 14 years, and Telfeian *et al*<sup>34</sup> reported recovery of language in a patient with hemispherectomy at age 16 years. McClelland and Maxwell<sup>37</sup> reported on nine adult patients, of whom three were employable and four leading a (semi-)independent life, and Schramm *et al*<sup>29</sup> found improved quality of life in 17/20 adults of our series.

Our data demonstrate that hemispheric surgery in eligible adolescent and adult patients can lead to favourable cognitive and psychosocial long-term outcomes, comparable or superior to those for ES patients. Importantly, these findings do not justify any suggestion to postpone hemispheric surgeries to a later brain maturation stage because the study was observational and can only retrospectively confirm the established surgical practice. Surgical decision making has and should rely on the comprehensive evaluation of several clinically relevant factors (eg, assumed progression and prospects), including age at surgery, and more severe or faster progressing epilepsies supposedly have been and will be submitted to surgery earlier than those less severe or progressing at a slower speed.<sup>5</sup> As was mentioned before, younger age at epilepsy onset, a lower presurgical cognitive level, more supportive training activities in early childhood, and the readiness for an early surgical intervention indicate earlier and more severe brain insults with stronger involvement of the contralateral hemisphere in ES patients. Thus, ES and LS patients are probably not the same type of patients and we cannot know how LS patients would have fared had they been operated on earlier, or ES patients when operated on later.

## Clinical prediction model

Our findings on the cognitive outcome seem to contradict influential neuropsychological intuitions. For example, higher age at surgery (ie, lower age-dependent neuroplasticity) was a positive predictor of postsurgical intelligence and cognitively high functioning patients showed the best overall outcome. Our data suggest a relatively simple non-dynamic model according to which the degree of functional integrity of the contralateral hemisphere determines the preoperative cognitive level and the postoperative cognitive potential. Evidently, functional shifts to the contralateral side (if any) must have been completed before hemispheric disconnection or resection. As the original insult in most patients occurred at a very young age, plasticity-based reorganisation will usually have been initiated long before surgery, rendering capacities for neural adaptation less important at the time of surgery. Higher age, instead, works as a positive, stabilising factor.<sup>18 39 40</sup> Aetiology and side of surgery did not further contribute to cognitive outcome prediction probably because their effects converge to the presurgical cognitive level. Furthermore, comparatively few patients with developmental and progressive aetiologies were included, which might have obscured potential differences. On top of the effects of the presurgical functional level and age at surgery, release from seizures (in combination with reducing or withdrawing AEDs) may increase the capacity of the remaining hemisphere resulting in cognitive improvements.

## Study limitations

Limitations of our study evolve from the use of postal questionnaires entailing possible response and recall biases as well as difficulties of the parents in evaluating a child's functional status. However, standardised neuropsychological assessment, particularly formal IQ testing, is typically not feasible or reliable in these patients.<sup>3 21</sup> In such cases, parental ratings constitute a useful psychometric tool of acceptable clinical validity. The high correlations between cognitive and psychosocial outcome actually confirm the adequacy and practical relevance of these data. Furthermore, parental ratings of presurgical intelligence correlated highly with presurgical IQ/developmental quotients for patients of whom data were available ( $r=0.695$ ;  $p<0.0001$ ;  $N=28$ ; postoperative data available for  $<20\%$  of patients). Comprehensive analyses on the medical outcome of the Bonn hemispherectomy series with further information on complications and aetiologies have recently been published.<sup>29 41</sup>

As stated above, truly experimental evaluations on the optimal timing of hemispheric surgeries are impossible. In our series, the age at surgery group factor was confounded by several factors, including presurgical intelligence. The study design also did not allow evaluating the preventive potential of hemispheric surgeries with regard to progressive cognitive-behavioural deterioration. Therefore, all analyses regarding age at surgery must be regarded as explorative. Supportive evidence from independent samples is required.

## CONCLUSION

In line with findings from other groups, our long-term outcome data corroborate the established clinical practice and the efficacy of hemispheric surgeries. Eligible older and cognitively high functioning patients can be perfect candidates for hemispheric surgery. Younger patients showed a minimally better seizure outcome. Patients with LS, however, exhibited better presurgical and postsurgical intelligence and, accordingly, achieved better academic and psychosocial outcomes. The overall cognitive and psychosocial long-term outcome is mainly determined by the

presurgical cognitive status, which represents an indicator of disease severity and the functional integrity of the contralateral hemisphere. Release from seizures is crucial for postsurgical cognitive improvement.

**Acknowledgements** We thank the patients and their families, who willingly gave their time and consent to support our research.

**Contributors** AA: analysis and interpretation of the data, drafting the article and revising it critically for important intellectual content, and final approval of the version to be published. UG: conception and design, analysis and interpretation of the data, and final approval of the version to be published. CH: conception and design, analysis and interpretation of the data, drafting the article and revising it critically for important intellectual content, and final approval of the version to be published. SB: analysis and interpretation of the data. The other authors contributed to revising the article critically for important intellectual content and gave final approval of the version to be published.

**Competing interests** None.

**Provenance and peer review** Not commissioned; externally peer reviewed

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